Exercise Sheet 1

Q1: Multiple Choice Questions

1. Which of the following best describes the relationship between Computer Graphics and Image Processing?
    a) Computer Graphics analyzes pictures to derive mathematical models; Image Processing creates images from models.
    b) Computer Graphics creates images from models; Image Processing analyzes pictures to derive descriptions.
    c) Both fields are identical in scope and application.
    d) Computer Graphics deals only with 3D models; Image Processing deals only with 2D photos.
2. Which input device type is best suited for interactive drawing, with the input primitive being a coordinate position (x, y)?
    a) String device
    b) Valuator
    c) Locator device
    d) Choice device
3. In a raster display, the frame buffer is used to:
    a) Store line-drawing instructions for vector graphics.
    b) Store the color value for each screen position (pixel).
    c) Convert Cartesian coordinates to normalized device coordinates.
    d) Perform clipping and rasterization.
4. Which of the following is a disadvantage of raster graphics compared to vector graphics?
    a) Raster graphics cannot display photographs.
    b) Enlarging a raster image causes pixelation or blurring.
    c) Raster graphics produce smaller file sizes.
    d) Raster graphics cannot use color blends.
5. In the OpenGL pipeline, which stage is responsible for creating pixels for each part of the object?
    a) Vertex Shader
    b) Primitive Assembly
    c) Clipping and Rasterization
6. Which of the following is NOT a component of the software level in interactive computer graphics?
    a) Application program
    b) Application model
    c) Graphics library (API)
    d) Frame buffer

Q2: True or False:

1. Vector displays store picture information as a set of line-drawing instructions.
2. Raster graphics produce smaller file sizes than vector graphics.
3. A valuator device provides scalar values typically between 0 and 1.
4. The frame buffer holds data representing all pixels in a complete video frame.
5. Transformations in 2D can be composed by multiplying matrices.
6. The vertex shader is a fixed-function stage in the OpenGL pipeline.
7. Vector graphics are best for images with complex color blends.
8. Clipping is the process of cutting vertices that will not appear on the screen.

Q3 Answer the following Questions.

1. Define Computer Graphics and list the three main operations it involves.
2. Differentiate between Geometric Models and Mathematical Models in computer graphics, giving one example of each.
3. Explain the difference between Non-interactive and Interactive computer graphics, providing one real-world example for each.
4. Describe the role of the GPU in the computer graphics system, particularly in relation to the CPU and frame buffer.
5. Briefly describe the main stages of the OpenGL pipeline.
6. Discuss the interdisciplinary nature of computer graphics. Explain how physics, mathematics, human perception, engineering, graphic design, and art each contribute to the field. Provide specific examples for each discipline.
7. Compare and contrast vector and raster display technologies. Your answer should address: How each constructs pictures, File size characteristics and Scalability and image quality
8. For each of the following projects, state whether vector or raster graphics is more appropriate and justify your choice:
    a. A company logo that will be printed on business cards and billboards.
    b. A digital photograph for a magazine cover.
    c. A simple icon for a mobile app interface.